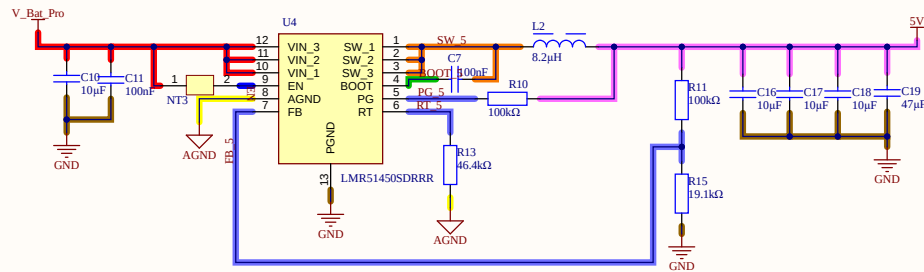
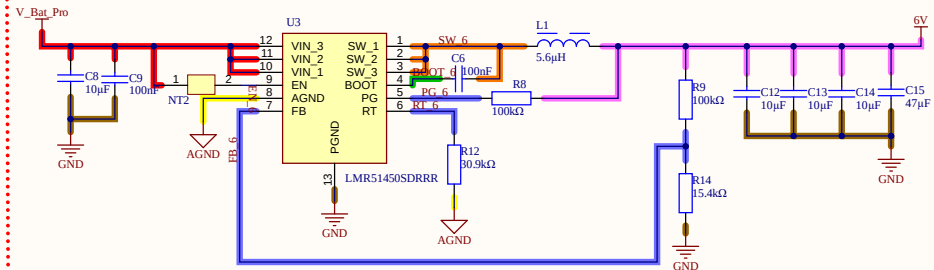


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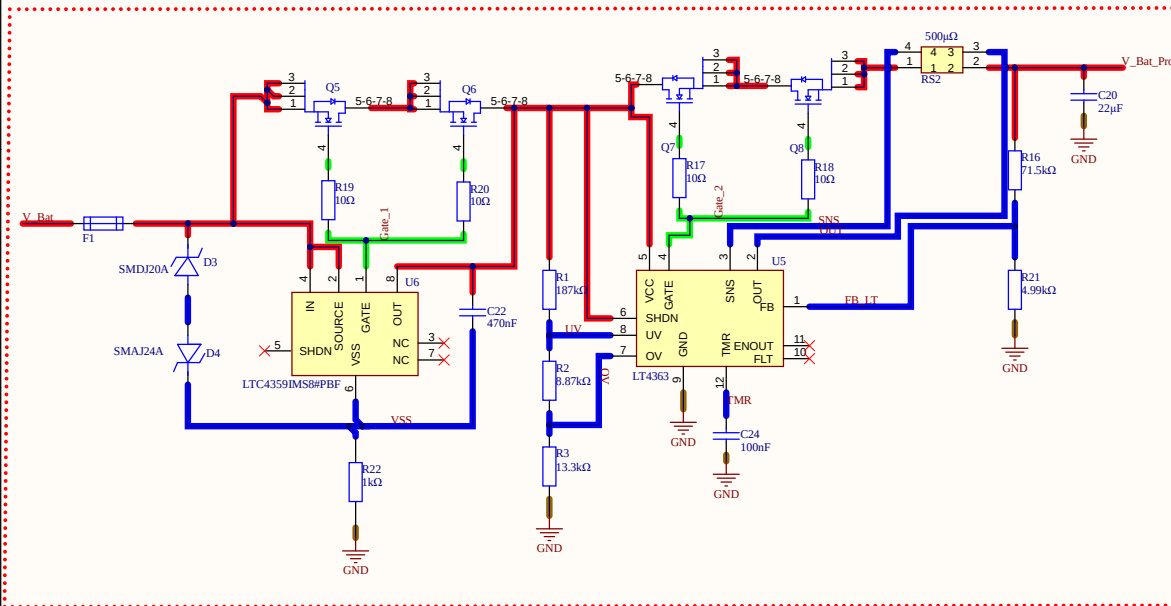
5V - 5A BUCK CONVERTER



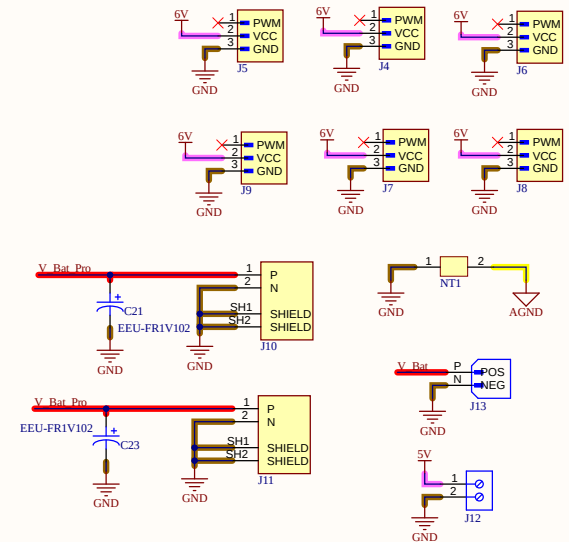
6V - 5A BUCK CONVERTER



PROTECTION



CONNECTORS AND NET-TIE



ORINIUM_UAV.PrjPcb

APPROVALS		DATE	
ENG:			
DSN:			
CHK:			
PROJECT REVISION:		DOCUMENT REVISION:	
Not in version control		Not in version control	
DESIGN ITEM:		VARIANT	
		[No Variations]	
REFERENCE DOCUMENTS		TITLE	
BOM:		"	
ASSY DWG:	SIZE	CAGE CODE	DWG NO.
FAB DWG:	B		
PCB DWG:	SCALE:	FILE NAME	REV
		PiAndServos.SchDoc	1

Altium

Altium Ltd.
4225 Executive Square
92037, La Jolla
USA

www.altium.com

1 of 1

A

Name	Description	Designator	Resident	Library	Quantity
GRM055F7C30R48D	50V 500pF X7R ±20% 1002 MultiLayer Ceramic Capacitors M.L.C.D. - 1806/3MT RCHS	CX, C7, C24	OP0402	GRM055F7C30R48D	3
C325C7R8-630R450AC	50V 220pF X7R ±20% 1220 MultiLayer Ceramic Capacitors M.L.C.D. - 1806/3MT RCHS	CX, C31, C32, C33, C34, C35, C37, C39	OP1220	C325C7R8-630R450AC	8
C100K040V07060004	MultiLayer Ceramic Capacitor, CC-301ms, 0.1 - 10%, 20%, X7S, 10 V, 0603 (1008 Metric)	CX, C31	OP0603	C100K040V07060004	2
GRM055F6C47R482L	50V 470pF X7R ±20% 1220 MultiLayer Ceramic Capacitors M.L.C.D. - 1806/3MT RCHS	C35, C39	OP1220	GRM055F6C47R482L	2
C375C7R8-630R450R8	50V 220pF X7R ±20% 1220 MultiLayer Ceramic Capacitors M.L.C.D. - 1806/3MT RCHS	C30	OP1220	C375C7R8-630R450R8	1
ELU1FR0V002	Radial Can ALUMINUM ELECTROLYTIC CAPACITORS 250V 1000µF ±20% 0.450mm 22.50mm	C21, C23	OPALUM1TH	ELU1FR0V002	2
W0F15C4739AT3F	FeedThru Ceramic Capacitors 120V 50V 4700pF	C22	OP0603	W0F15C4739AT3F	1
9M4J50A	9M4J Series 20V/3000 W/Surface Mount UHF Directional TVCAs - DO-214AB	C3	9M4C(DO-214AB)	9M4J50A	1
9M4J24A	9M4J Series 400V/24 V-H-Directional Surface Mount TVS Diode - DO-214AC	C4	9M4C(DO-214AC)	9M4J24A	1
9F-3812P000T-2	9F-3812Pac, Res Acting: 60A, 250VAC, 120VDC, 3812 (10003 Metric), 9F-3812P-T	F1	9F-3812P000T-2	9F-3812P000T-2	1
6L30031121	WHP40 Straight Male, PCB Header Through Hole, 1 Row, 10 Contacts, 2.54mm Pitch	J1, J5, J6, J7, J8, J9	LX3 MALE16P00R	6L30031121	6
X70PQ4M	X70PQ4M	B1, B1	X70PQ4M	X70PQ4M	2
175066	Black, Terminate: 6A, 60 V, 2.54mm, 36 W 20 AWG, 2, 5mm, Soldering 1.5 AW	B2	175066	175066	1
X70S5	X70S5	B3	X70S	X70S5	3
VAL000-620MB	General Purpose Inductor, 5.6H, 20%, 1 Element, Composite Core, SMD, 2525	L1	VAL000-620MB	VAL000-620MB	1
VAL000-620MB	VAL000-620MB	L2	VAL000-620MB	VAL000-620MB	3
BES120N04LS	Transistor, N- MOSFET, unidirect, 40V, 20W, 120mVr, 6 75C/W, 6	Q5, Q6, Q7, Q8	80N 046	BES120N04LS	4
PCD05FR17257N1	Res Thick Film 0605 100K Ohm 2% 0.125W (1/8W) ±50ppm/°C Chd SMD TFR	R1	RES1206	PCD05FR17257N1	1
PCD05FR17892L1	Res Thick Film 0605 100K Ohm 2% 0.125W (1/8W) ±50ppm/°C Chd SMD TFR	R2	RES1206	PCD05FR17892L1	1
PCD05FR1713K1L	Res Thick Film 0605 11.3K Ohm 2% 0.125W (1/8W) ±50ppm/°C Chd SMD TFR	R3	RES1206	PCD05FR1713K1L	1
CHCM0032300P48A	120mW Thick Film Resistor 75V ±50ppm/°C, ±2% 10000/1000 Chip Resistor - Surface Mount RCHS	RL, R20	RES0603	CHCM0032300P48A	2
CHCM0422300P48D	320mW Thick Film Resistor 75V ±2%, ±50ppm/°C, 10000/ 1000 Chip Resistor - Surface Mount RCHS	RL, RL1	RES0402	CHCM0422300P48D	2
CHCM0422300P48D	Thin Film Resistor, Metal Glass Thick Film, 0.50W, 10000WV, 100V, 2% ± Tol, Surface Mount, 0603	R2	RES0402	CHCM0422300P48D	1
CHCM0032400P48A	Thin Film Resistor, Metal Glass Thick Film, 0.1W, 4000WV, 75V, 2% ± Tol, Surface Mount, 0603	R23	RES0603	CHCM0032400P48A	1
ACD05FR17257N1L	Res Thick Film 0402 15.4K Ohm 2% 0.100W (1/10W) ±50ppm/°C Chd SMD Automotive TFR	R24	RES0402	ACD05FR17257N1L	1
CHCM0422300P48D	Res Thick Film 0402 10.1K Ohm 2% 1/10W ±50ppm/°C Chd SMD Automotive TFR	R25	RES0402	CHCM0422300P48D	1
PCD05FR17725L1	Res Thick Film 0603 71.5K Ohm 2% 0.1W (1/10W) ±50ppm/°C Chd SMD TFR	R26	RES0603	PCD05FR17725L1	1
9F0805FR17002L	Res Thick Film 0605 10 Ohm 2% 0.25W (1/4W) ±50ppm/°C Chd SMD Automotive TFR	R27, R28, R29, R30	9F0805FR17002L	9F0805FR17002L	4
9F039F4805X	Res Thick Film 0402 4.75K Ohm 2% 0.1W (1/10W) ±50ppm/°C Chd SMD Automotive TFR	R21	RES0402	9F039F4805X	1
9F120FR17725L	Thin Film Resistor, Metal Glass Thick Film, 0.25W, 10000WV, 100V, 2% ± Tol, Surface Mount, 1008	R22	RES1206	9F120FR17725L	1
W6P420L5300PFA	Res Metal 3p 4205 0.0005 Ohm 2% 1W ±75ppm/°C Metal Pkd SMD Automotive TFR	R32	9P1_4205	W6P420L5300PFA	1
UAP614020EPRR	30 V, 5.6 A synchronous buck converter 25VDC/4 40 to 150	U1, U4	MECN-12	UAP614020EPRR	2
U140B3	Circuit Protection Misc High Voltage Surge Suppressor with Current Limit	U5	MECP-12	U140B3MS21P0F	1
U1C400M0808P0F	CMF Controller N- Channel Single Automotive P-MOSFET (MECP Tube) / IC CR CMF400M0808P0F	U6	MECP-8	U1C400M0808P0F	1